Liz Motor Corp.: Capital Budgeting for an ESG Project

Ivey Case W32397 — Answers to the Assigned Questions

Question 1 — Qualitative Assessment: Is Liz Well Positioned for Success?

Strengths

- Liz has aspired to EV market leadership since its 2012 founding and has expanded from cars into battery-powered buses and trucks, with healthy sales growth over the past four years driven by younger consumers' interest in its compact and sports cars.
- As a young company, Liz is structurally lighter than legacy automakers: it has no extensive dealer network, no entrenched labour unions, and no legacy fuel-powered business it must protect while transitioning — incumbents can't simply walk away from the business that still generates most of their revenue.
- Liz is investing heavily in a modular EV production platform across factories in the US, Canada, and China, and has secured $3.5 billion in financing over the next three years to launch 20 new EV models by 2027.
- Liz surprised the market by announcing it will manufacture its own batteries for its entire EV line-up — vertical integration that, combined with first-mover use of solid-state battery technology, could be a genuine differentiator (a 39% cut in battery-related carbon footprint) at large scale, ahead of any major competitor.
- Regulatory tailwinds favour EV investment broadly: the US has pledged to halve greenhouse-gas emissions by 2030, the EU has proposed a 55% CO2 cut by 2030 (and a further 100% cut over the following five years, making fossil-fuel vehicle sales in the EU nearly impossible by 2035), and China has pledged carbon neutrality by 2060.

Weaknesses / Risks

- Liz's sales growth is decelerating sharply — from 26% in 2021 to 17% in 2022 — as the EV market becomes crowded with legacy automakers adding EVs to their line-ups, eroding Liz's early advantage.
- Liz's ESG rankings for 2022 were weak enough to drop the stock price 6.2% in a single reaction, prompting the board to demand a 'quick turnaround' — ESG has become a live strategic vulnerability, not a side issue.
- The $3.5 billion financing commitment plus this project's own $800M+ up-front outlay represent very high capital intensity and execution risk, and solid-state battery technology is unproven at scale — Liz would be the first major automaker to deploy it, with no track record to rely on.
- Liz is exposed to interest-rate risk (a Fed rate hike raises its cost of debt, as Scenario (a) shows) and competitive/technology risk (rival automakers or a hydrogen breakthrough could erode the project's returns, as Scenario (f) shows).

Assessment: Liz is reasonably well positioned, but not comfortably so. Its structural advantages over legacy automakers (no legacy business to protect, secured financing, vertical integration into batteries) and the regulatory tailwind behind EVs are real. But the two things that made Liz distinctive — fast growth and a clean ESG story — are both under active pressure: growth is decelerating as the market crowds, and the 2022 ESG rankings triggered a genuine market reaction. The solid-state battery project is a direct response to both problems at once, which is exactly why it is strategically significant even before the DCF numbers are examined.

Question 2 — What Hurdle Rate Should Be Used?

Anthony is financing this specific project with debt, and the case states its after-tax cost of debt directly: 8% in the base case (falling to 7.5% with a green bond, or rising to 8.5% if the Fed keeps raising rates). Because the case ties this project's financing to debt and gives no information about Liz's cost of equity, beta, or target capital structure,

the after-tax cost of debt — 8% — is the hurdle rate used throughout this analysis, consistent with how the case itself frames it ("the cash flows and the hurdle rate given above...").

A caveat worth flagging to the board: per the WACC methodology in the course material, WACC = (E/V) × Ke + (D/V) × Kd × (1 − T) is the theoretically correct discount rate for a project of average firm risk, and equity is virtually always priced above the cost of debt. If Liz does not finance its overall capital budget entirely with debt, using only the after-tax cost of debt as the hurdle rate is a conservative, project-specific choice — it may understate the true risk-adjusted return the project needs to clear. The 8% (7.5%–8.5%) figures used below should be read as the rate for this debt-financed project specifically, not as Liz's corporate WACC.

Question 3 — Base Model: Cash Flows, NPV, Payback, IRR, PI

Exhibit 1 of the case gives the cash-flow timeline for 2023–2035. Treating 2023 as time 0 (the investment decision date):

Year	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
Period (t)	0	1	2	3	4	5	6	7	8	9	10	11	12
Cash flow ($M)	-800	-20	-20	-20	-20	200	200	200	200	200	200	200	200

The $800M in 2023 is the technology-acquisition outlay; $20M/year for the next four years (2024–2027) develops and adopts it; returns of $200M/year begin in the fifth year of the technology's life (2028) and run for eight more years, consistent with its stated 12-year life (12 years of post-acquisition cash flows, 2024–2035).

Metric	Value	Formula used
NPV	−$21.45 million	NPV = Σ CFt /(1+r)^t
IRR	7.66%	Rate that sets NPV = 0
PV of inflows	$844.79 million	Σ (inflows)/(1+r)^t
PV of outflows	$866.24 million	Σ (outflows)/(1+r)^t
Profitability Index (PI)	0.975	PI = PV(inflows) / PV(outflows)
Payback period (undiscounted)	8.40 years	Years to recover the initial outlay
Discounted payback period	Not recovered within the 12-year life	Same, using discounted cash flows

Reading: at the 8% after-tax cost of debt, the base case narrowly destroys value. NPV is negative, IRR (7.66%) sits just below the 8% hurdle, PI is just under 1.0, and the project never fully repays its cost on a discounted basis within its 12-year life. On a pure DCF basis this is a marginal "reject" — which is exactly why the sensitivity analysis in Question 4 matters so much.

Question 4 — Sensitivity Analysis

Each of the six uncertainties listed in the assignment is analyzed by changing exactly the input the case describes and recomputing NPV at the base 8% rate (unless the uncertainty is itself a rate change).

(a) Fed continues raising rates: after-tax Kd = 8.5%

Metric	Base Case	Scenario (a)
NPV	−$21.45M	−$51.69M
Δ NPV	—	−$30.24M

A higher cost of debt makes an already-marginal project meaningfully worse — NPV falls by over $30M.

Variable changed and why: the discount rate — a Fed rate hike is a macroeconomic, interest-rate risk entirely outside Liz's control. Isolating it (holding cash flows fixed) shows how sensitive the project is to the cost of debt alone.

Business impact: NPV falls from −$21.45M to −$51.69M — a $30.24M swing from a single external rate move.

(b) Green bond financing: after-tax Kd = 7.5%

Metric	Base Case	Scenario (b)
NPV	−$21.45M	+$10.20M
Δ NPV	—	+$31.65M

This is the single most important lever in the analysis: switching to green-bond financing, on its own, flips the project from value-destroying to value-creating.

Variable changed and why: the discount rate again, but this time from a financing choice Liz controls directly (green bond vs. ordinary debt), not an external shock.

Business impact: NPV moves from −$21.45M to +$10.20M — a $31.65M swing that, on its own, flips the accept/reject decision.

(c) Carbon Border Adjustment Mechanism (CBAM): inflows rise to $220M/year

Metric	Base Case	Scenario (c)
NPV	−$21.45M	+$63.03M
Δ NPV	—	+$84.48M

A 10% increase in the annual inflow swings NPV from negative to strongly positive — the project is highly leveraged to the size of the annual cash flow.

Variable changed and why: the annual cash inflow, because the EU's CBAM policy is an external regulatory tailwind that raises the revenue/margin benefit of Liz's solid-state batteries.

Business impact: a 10% rise in the annual inflow swings NPV from −$21.45M to +$63.03M — an $84.48M change.

(d) Sale of the technology at the end of year 12 for $100M

Metric	Base Case	Scenario (d)
NPV	−$21.45M	+$18.26M
Δ NPV	—	+$39.71M

Even a distant, deeply-discounted salvage value is enough to turn the project NPV-positive, since the base case was only marginally negative.

Variable changed and why: a one-time terminal cash flow, representing the option to monetize the technology by selling it to another automaker instead of only running it to the end of its life.

Business impact: NPV moves from −$21.45M to +$18.26M — a $39.71M swing, even though the $100M is 12 years out and heavily discounted.

(e) Technology more valuable than expected: $210M/year, or a $810M sale today

Metric	Value
NPV at $210M/year (r = 8%)	+$20.79M
Δ NPV vs. base case	+$42.24M
PV of the inflow stream alone (the technology's DCF	$887.03M

value)	
Quoted market value of the technology today (2023), per the case	$810.00M

Cross-checking the two valuations (the same way we cross-checked ITC's DCF value against its market price in the WACC assignment): the DCF value of the technology's future cash flows ($887M) exceeds the market's quoted value ($810M). That gap suggests either the market is pricing the technology conservatively, or the $210M/year assumption is somewhat optimistic — either way, this scenario is comfortably NPV-positive under either valuation.

Variable changed and why: the annual cash inflow again, this time reflecting technology-performance/demand upside rather than a policy change.

Business impact: NPV moves from −$21.45M to +$20.79M — a $42.24M swing — and the DCF value cross-checks reasonably closely against the case's independent $810M market valuation.

(f) Hydrogen breakthrough risk: 30% chance the project life shortens to 10 years, salvage wiped out

If hydrogen vehicles dominate, the project's life shortens from 12 to 10 years post-acquisition (cash flows run 2024–2033 instead of 2024–2035, losing the last two $200M inflows), with 30% probability. The expected NPV blends this outcome with the 70% chance the base case holds (Expected value = Σ Probability × Outcome).

Metric	Value
NPV if life shortens to 10 years	−$186.65M
NPV if base case (12-year life) holds	−$21.45M
Expected NPV = 0.30 × (−186.65) + 0.70 × (−21.45)	−$71.01M
Δ Expected NPV vs. base case	−$49.56M

Losing the final two years of $200M inflows is costly even discounted back, and a 30% chance of it happening pulls the expected NPV down to −$71M — this is the most damaging of the six scenarios.

Variable changed and why: the project's life (and therefore how many $200M inflows are ever received), because a hydrogen breakthrough is an external technology-obsolescence risk, not something Liz controls.

Business impact: weighted by its 30% probability, expected NPV falls from −$21.45M to −$71.01M — a $49.56M reduction, the largest quantified external threat in the analysis.

Summary of the Sensitivity Analysis (r = 8% unless noted)

#	Scenario	NPV ($M)	Δ NPV vs. base ($M)
Base	Base case (8% hurdle rate)	−21.45	—
(a)	Fed hikes rates → Kd = 8.5%	−51.69	−30.24
(b)	Green bond → Kd = 7.5%	+10.20	+31.65
(c)	CBAM tailwind → $220M/yr	+63.03	+84.48
(d)	Sale of tech in yr 12 for $100M	+18.26	+39.71
(e)	More valuable tech → $210M/yr	+20.79	+42.24
(f)	30% chance of 10-yr life, expected value	−71.01	−49.56

Why Each Variable Was Changed, and Its Business Impact

Each scenario is a one-variable-at-a-time (ceteris paribus) sensitivity test: everything else is held at the base case so the effect of that one uncertainty can be seen in isolation. This is standard capital-budgeting sensitivity analysis, per the Project Evaluation Methods PPT, and it tells the board exactly which single lever or risk drives how much of the outcome.

Scenario	Variable changed	Business rationale / impact
Base case	—	Reference point every scenario below is measured

		against.
(a) Fed hikes rates	Discount rate	External macro/interest-rate risk, outside Liz's control. Shows sensitivity to cost of debt alone.
(b) Green bond	Discount rate	Financing choice within Liz's control. The single biggest positive lever — flips reject to accept.
(c) CBAM tailwind	Annual cash inflow	External regulatory tailwind (EU policy). Shows how leveraged NPV is to the annual cash flow size.
(d) Sale of tech (yr 12)	Terminal cash flow	Exit/monetization option. Shows the technology has asset value beyond its operating cash flows.
(e) More valuable tech	Annual cash inflow	Execution/demand upside. Cross-checked against an independent $810M market valuation.
(f) Hydrogen breakthrough	Project life (# of inflows)	External technology-obsolescence risk. The largest quantified external threat, weighted by its 30% probability.

Question 5 — Recommendation to the CEO and Board

On a strict, base-case DCF test, the project is a marginal reject: NPV is slightly negative (−$21.45M), IRR (7.66%) sits just under the 8% hurdle rate, and PI is just under 1.0. The sensitivity analysis shows this conclusion is fragile in both directions:

- The financing decision alone can flip the outcome — moving from ordinary debt to a green bond (Scenario b) turns NPV positive by over $30M, and that choice is largely within Liz's control.
- Regulatory tailwinds already in motion (CBAM, Scenario c) and potential technology upside (Scenarios d and e) all push NPV comfortably positive.
- The main quantified downside risk — a 30% chance of technology obsolescence from a hydrogen breakthrough (Scenario f) — pulls expected NPV to about −$71M, and is largely outside Liz's control.

Beyond the DCF, the case also lists ESG benefits that cannot be quantified in the model but are directly relevant given Liz's weak 2022 ESG rankings: building social capital and trust (potentially lowering the firm's own cost of capital), improving corporate image and brand value, product differentiation that reduces demand elasticity and risk, cost savings from reduced waste and avoided carbon-pricing exposure, reduced regulatory risk, possible future EV-buyer incentives, and improved employee retention and motivation.

Recommendation: proceed with the project, but make green-bond financing a condition of approval rather than a fallback — on these numbers, it is what turns a marginal, DCF-negative project into a value-creating one. Given how close the base case sits to zero, the unquantified ESG benefits are not a footnote; they are the deciding factor that should tip a near coin-flip financial case toward acceptance, particularly given the board's explicit demand for a 2023 ESG turnaround. Anthony should present the board with the full range of outcomes above, not a single point estimate, and flag green-bond financing (b), the CBAM tailwind (c), and the hydrogen-breakthrough risk (f) as the three factors most worth acting on or actively monitoring.

Formulas Used in This Analysis

Every formula below is the one actually applied above. None has been introduced from outside the course PPTs / formula notes.

Formula	Used for
NPV = Σ CFt / (1+r)^t	Base model, and Scenarios (a), (b), (c), (d), (e), (f) — recomputing NPV under a changed rate or changed cash flow
IRR = the discount rate that sets NPV = 0	Base model — cross-checking NPV against the 8% hurdle rate
Profitability Index, PI = PV(inflows) / PV(outflows)	Base model
Payback period = years to recover the initial outlay (undiscounted cash flows)	Base model

Discounted payback period = same, using discounted cash flows	Base model
Expected value = Σ (Probability × Outcome)	Scenario (f) — blending the 30%-probability shortened-life NPV with the 70%-probability base-case NPV
WACC = (E/V) × Ke + (D/V) × Kd × (1 − T)	Question 2 — noted as the theoretically correct hurdle rate, for context on why the after-tax cost of debt is used as a project-specific simplification

Appendix — Additional Scenarios Beyond the Assigned Questions

The case describes four further uncertainties that are not part of the assigned questions (competitor entry, the abandonment option, gradual market acceptance, and inflation). They are included here for completeness, using the same 8% base hurdle rate unless noted.

Extra scenario	NPV ($M)	Δ NPV vs. base ($M)	Note
Competitor entry by 2027 → inflows fall to $180M/yr	−105.93	−84.48	Worst of all scenarios in the case
Abandon in 2028 for $500M vs. continue to 2035	−21.45 (continue optimal)	0.00	PV of continuing ($1,241.27M, annuity due) far exceeds the $500M offer, so the option is not exercised
Gradual market acceptance: $152M in 2028, growing 8%/yr	−38.65	−17.20	r = g = 8%, so the growing-annuity formula is undefined; cash flows discounted term-by-term instead
3% inflation, treated consistently (Fisher effect)	−21.45	0.00	Nominal rate = 1.08×1.03−1 = 11.24%; inflating cash flows and raising the rate consistently leaves NPV unchanged

Why Each Variable Was Changed, and Its Business Impact (Appendix Scenarios)

Scenario	Variable changed	Business rationale / impact
Competitor entry	Annual cash inflow	External competitive risk (mirror image of Scenario c). Worst outcome in the whole case.
Abandonment option	Continue-vs-exit decision	Real option value. Not exercised at these numbers — continuing is worth far more than the $500M offer.
Gradual acceptance	Timing/shape of inflows	Delaying value to later years hurts NPV even if cash flows eventually exceed the base case.
Inflation (Fisher)	Cash flows & discount rate (jointly)	Tests whether inflation itself matters. It doesn't, when cash flows and the rate are inflated consistently.

Note on methodology: all discounting uses the after-tax cost of debt (8%) as the hurdle rate, per the case's own statement that this rate funded the project (see Question 2 for a discussion of this choice).